

Trigonometric Functions

9

SYLLABUS 3.5–3.11

Watch a single seat on a turning Ferris wheel and forget the rest of the ride. As the wheel turns steadily, that seat rises, slows near the top, falls, slows near the bottom, and climbs again, over and over, forever the same. Plot its height against time and you get a smooth, endlessly repeating wave.

That wave is a **sine curve**, and it is no coincidence. The seat is a point moving round a circle at constant speed; its height is just the vertical coordinate of that point. The circle and the wave are the same motion seen two ways, one going round, one going along.

This is the hinge of the whole chapter. In Ch. 8 the sine and cosine were ratios inside a triangle, trapped between 0 and 90° . Freed onto the circle, they become functions of *any* angle, and functions that repeat. From that single shift, everything follows: exact values you can read off the circle, identities that relate the functions to one another, graphs that oscillate without end, and equations whose solutions come not one at a time but in endless families.

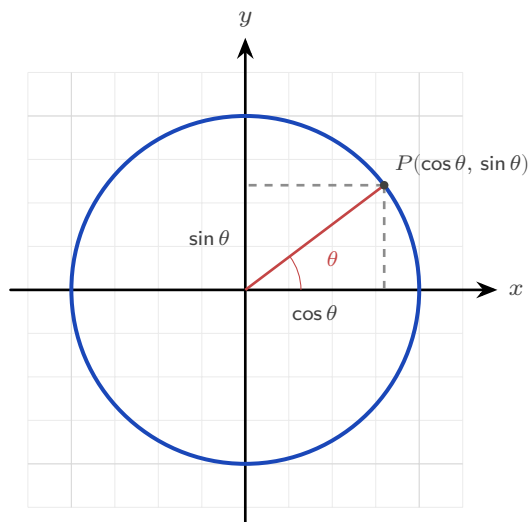
Oscillation is everywhere the moment you look: tides, seasons, sound, light, alternating current, the swing of a pendulum, the beat of a heart. All of them are, at bottom, a point going round a circle. Learn the circle and you have learned them all.

9.1 The Unit Circle

Take the circle of radius 1 centred at the origin, and a point P on it at angle θ measured anticlockwise from the positive x -axis. The old right-triangle ratios still hold, but now the hypotenuse is 1, so the two shorter sides *are* the coordinates of P . This turns a definition trapped in acute angles into one that works for every real θ .

DEF On the unit circle, for a point P at angle θ :

$$\cos \theta = x\text{-coordinate of } P, \quad \sin \theta = y\text{-coordinate of } P.$$



The tangent joins them as the gradient of the line OP , so $\tan \theta = \frac{\text{rise}}{\text{run}} = \frac{\sin \theta}{\cos \theta}$, undefined wherever $\cos \theta = 0$.

KEY · IB $\tan \theta = \frac{\sin \theta}{\cos \theta}.$

Exact values

A few angles give exact coordinates, from the 30–60–90 and 45–45–90 triangles. They are worth knowing cold, because they turn up on every non-calculator paper.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	—

Symmetry and the sign of each function

The four quadrants of the circle share coordinates up to sign, so the value of a trig function at any angle is the value at a related acute angle, with a sign fixed by the quadrant. The mnemonic **CAST** records which functions are positive: **A**ll in the first quadrant, then **S**ine,

Tangent, Cosine going anticlockwise. From the reflection symmetries of the circle,

$$\sin(\pi - \theta) = \sin \theta, \quad \cos(-\theta) = \cos \theta, \quad \sin(\theta + 2\pi) = \sin \theta,$$

and similar relations for every reflection and full turn. Because a full revolution returns P to its start, both functions repeat every 2π : they are **periodic**.

The first of these relations is an old acquaintance. Two angles, θ and $\pi - \theta$, share the same sine, and that shared sine is exactly what produced the ambiguous case of the sine rule in Ch. 8: a triangle built from a sine value can sometimes close in two different ways, one acute and one obtuse.

YOUR TURN 9.1 The Unit Circle

1. State the exact value of $\sin \frac{\pi}{6}$. [1]
2. State the exact value of $\cos \frac{\pi}{4}$. [1]
3. State the exact value of $\tan \frac{\pi}{3}$. [1]
4. State the exact value of $\sin \frac{\pi}{2}$. [1]
5. Use symmetry to find the exact value of $\sin \frac{5\pi}{6}$. [2]
6. Use symmetry to find the exact value of $\cos \frac{2\pi}{3}$. [2]
7. In which two quadrants is $\sin \theta$ positive? [1]

9.2 Trigonometric Identities

An **identity** is an equation true for every angle. The most fundamental one is Pythagoras' theorem read off the unit circle: the point $(\cos \theta, \sin \theta)$ is distance 1 from the origin, so $\cos^2 \theta + \sin^2 \theta = 1$.

KEY · IB $\cos^2 \theta + \sin^2 \theta = 1$.

Double and compound angles

The functions of a sum of two angles can be expanded, and from the sum formulas everything else follows. These **compound-angle** identities, an HL tool, are proved from the geometry of two stacked triangles, but they are used as machinery.

KEY · IB **HL** **Compound angles:**

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B, \quad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B,$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}.$$

Setting $B = A$ collapses each of these into a **double-angle** identity, the single most useful family in the topic. The sine and cosine versions sit in the SL half of the booklet; the tangent version is HL.

KEY · IB **Double angles:**

$$\sin 2\theta = 2 \sin \theta \cos \theta, \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta.$$

KEY · IB **HL** $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$

The three forms of $\cos 2\theta$ are the same identity dressed differently by $\sin^2 + \cos^2 = 1$; you pick whichever leaves you with only sines or only cosines, which is exactly what makes it so useful when solving equations.

EXAMPLE 9.1

Find the exact value of $\sin 75^\circ$.

Write $75^\circ = 45^\circ + 30^\circ$ and use the compound-angle formula:

$$\sin 75^\circ = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Splitting an awkward angle into a sum of two known ones is the standard trick for exact values beyond the basic table.

Identities also let you move between ratios *without ever finding the angle*. This is a favourite examination move: the angle stays unknown from start to finish, and the Pythagorean identity does all the work.

EXAMPLE 9.2

Given that $\cos x = \frac{3}{4}$ and x is acute, find the exact value of $\sin 2x$ without finding x .

From $\cos^2 x + \sin^2 x = 1$, $\sin^2 x = 1 - \frac{9}{16} = \frac{7}{16}$, and since x is acute, $\sin x = \frac{\sqrt{7}}{4}$ (positive).

Then

$$\sin 2x = 2 \sin x \cos x = 2 \cdot \frac{\sqrt{7}}{4} \cdot \frac{3}{4} = \frac{3\sqrt{7}}{8}.$$

The word “acute” fixed the sign of $\sin x$. If x had instead been in the fourth quadrant, the same working would give $\sin 2x = -\frac{3\sqrt{7}}{8}$: always let the quadrant choose the sign before doubling.

Proving an identity is the same game played in reverse: start from one side, substitute known identities, and simplify until the other side appears.

EXAMPLE 9.3

Prove that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$.

Work on the left side, choosing the form $\cos 2\theta = 2 \cos^2 \theta - 1$ because it swallows the 1:

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{2 \sin \theta \cos \theta}{1 + (2 \cos^2 \theta - 1)} = \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta} = \frac{\sin \theta}{\cos \theta} = \tan \theta. \quad \blacksquare$$

Choosing the form of $\cos 2\theta$ that cancels the constant is the same instinct as in equation solving: pick the version that matches what sits next to it.

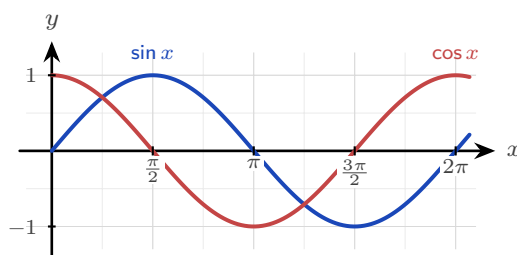
YOUR TURN 9.2 Trigonometric Identities

8. Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$. [2]
9. Simplify $\sin^2 x + \cos^2 x$. [1]
10. Write $\sin 2\theta$ in terms of $\sin \theta$ and $\cos \theta$. [1]
11. Given $\sin \theta = 0.6$ and $\cos \theta = 0.8$, find $\sin 2\theta$. [2]
12. Simplify $2 \cos^2 \theta - 1$. [1]
13. Expand $\sin(A + B)$. [1]
14. Find the exact value of $\cos 15^\circ$ using a compound angle. [3]

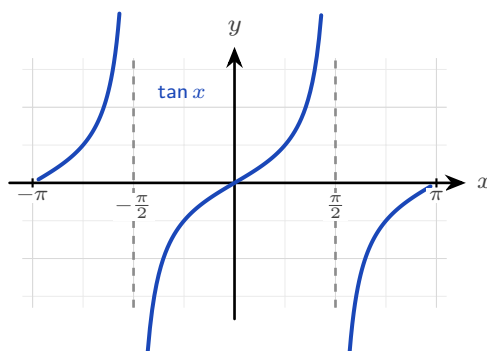
9.3 Graphs of Trigonometric Functions

Unrolling the point's motion around the circle onto a time axis gives the graphs. Because the motion repeats every 2π , so do $y = \sin x$ and $y = \cos x$: smooth waves of amplitude 1,

the cosine being the sine shifted left by $\frac{\pi}{2}$.



The tangent behaves differently. It has period π , not 2π , and because it is $\frac{\sin}{\cos}$ it shoots to $\pm\infty$ wherever $\cos x = 0$, giving vertical asymptotes at $x = \frac{\pi}{2} + n\pi$.



Transforming the wave

The general sinusoid stretches and slides the basic sine, and each parameter has a plain meaning.

KEY For $y = a \sin(b(x + c)) + d$:

amplitude = $|a|$, period = $\frac{2\pi}{|b|}$, horizontal shift = $-c$, vertical shift (midline) = d .

EXAMPLE 9.4

Describe $y = 3 \sin(2x) - 1$ and state its range.

The amplitude is 3, so the wave rises and falls 3 either side of its midline. The period is $\frac{2\pi}{2} = \pi$, so it completes a full cycle twice as fast as $\sin x$. The midline is $y = -1$. The graph therefore runs from $-1 - 3 = -4$ up to $-1 + 3 = 2$, giving range $-4 \leq y \leq 2$.

The four parameters are not just labels on a graph; they are how a sinusoid is fitted to the real world. The opening story can now be settled with actual numbers.

EXAMPLE 9.5

When it opened in 1999, the Daikanransha in Odaiba, Tokyo, was the world's tallest Ferris wheel: 115 m tall, its wheel 100 m across, turning once every 16 minutes. A seat starts at the bottom

of the wheel. Write the seat's height $h(t)$ in metres after t minutes, and find its height after 5 minutes.

Read each parameter off the wheel. The axle sits at $115 - 50 = 65$ m, so the midline is $d = 65$. The radius gives the amplitude, $a = 50$. One revolution takes 16 minutes, so $b = \frac{2\pi}{16} = \frac{\pi}{8}$. The seat starts at the *bottom*, 15 m up, which is what an upside-down cosine does at $t = 0$:

$$h(t) = 65 - 50 \cos\left(\frac{\pi t}{8}\right).$$

Check the skeleton before trusting it: $h(0) = 65 - 50 = 15$ (bottom) and $h(8) = 65 + 50 = 115$ (top, half a turn later). Then

$$h(5) = 65 - 50 \cos \frac{5\pi}{8} \approx 65 + 19.1 \approx 84.1 \text{ m.}$$

Choosing $-\cos$ to start at the minimum, rather than forcing a phase shift into a sine, is the modelling move examiners reward: match the *shape* of the motion first, then the numbers.

Symmetry of the graphs HL

Each reflection symmetry of the circle from Section 9.1 becomes a visible symmetry of the graphs. The cosine curve is mirror-symmetric in the y -axis, so $\cos(-x) = \cos x$: cosine is an **even** function. The sine and tangent curves map onto themselves under a half-turn about the origin, so $\sin(-x) = -\sin x$ and $\tan(-x) = -\tan x$: both are **odd**, in the sense of Ch. 5. The same circle symmetries give the supplementary-angle relations

$$\sin(\pi - \theta) = \sin \theta, \quad \cos(\pi - \theta) = -\cos \theta, \quad \tan(\pi - \theta) = -\tan \theta,$$

which you can see on the graphs directly: the sine curve is symmetric about the vertical line $x = \frac{\pi}{2}$, while cosine and tangent flip sign across it. In an exam these relations save you from ever reaching for a calculator: $\cos \frac{3\pi}{4} = -\cos \frac{\pi}{4} = -\frac{\sqrt{2}}{2}$ in one step.

YOUR TURN 9.3 Graphs of Trigonometric Functions

15. State the period of $y = \sin x$. [1]
16. State the amplitude of $y = 3 \cos x$. [1]
17. State the period of $y = \sin 2x$. [1]
18. State the range of $y = 2 \sin x + 1$. [2]
19. State the period of $y = \tan x$. [1]
20. State the equations of two asymptotes of $y = \tan x$. [2]

21. State the amplitude, period and midline of $y = 4 \sin(3x) - 2$.

[3]

9.4 Reciprocal and Inverse Functions HL

At Higher Level three more functions join the family, the reciprocals of the first three.

$$\text{KEY} \cdot \text{IB} \quad \sec \theta = \frac{1}{\cos \theta}, \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta}.$$

The third, the **cotangent**, is $\cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$; its definition is not printed in the booklet, so it is worth a moment of memory. Each reciprocal function blows up where its parent function vanishes: $\sec$ has asymptotes where $\cos \theta = 0$, while cosec and $\cot$ have them where $\sin \theta = 0$.

Dividing the Pythagorean identity $\cos^2 \theta + \sin^2 \theta = 1$ through by $\cos^2 \theta$, then by $\sin^2 \theta$, produces two more identities that convert between the reciprocal functions.

$$\text{KEY} \cdot \text{IB} \quad 1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta.$$

These identities exist to turn mixed equations into single-function ones, exactly as the double angles did.

EXAMPLE 9.6

Solve $\tan^2 x + \sec x = 1$ for $0 \leq x \leq 2\pi$.

The equation mixes $\tan$ and $\sec$; the identity $\tan^2 x = \sec^2 x - 1$ rewrites it in $\sec x$ alone:

$$\sec^2 x - 1 + \sec x = 1 \implies \sec^2 x + \sec x - 2 = 0 \implies (\sec x + 2)(\sec x - 1) = 0.$$

So $\sec x = -2$ or $\sec x = 1$, which is $\cos x = -\frac{1}{2}$ or $\cos x = 1$. From the circle,

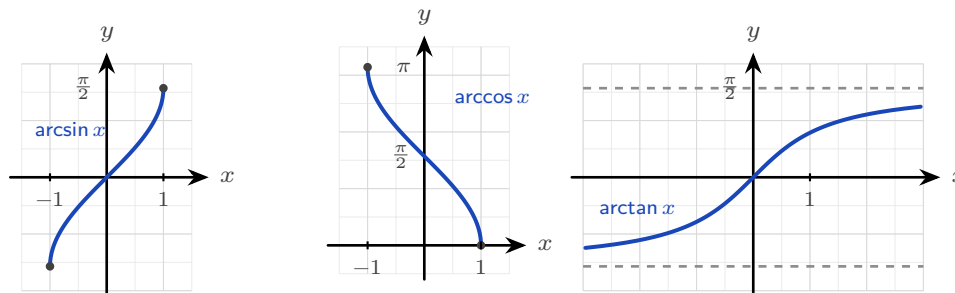
$$x = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \text{or} \quad x = 0, 2\pi.$$

Converting back to $\cos$ at the end is the safest final step: the unit circle speaks sine and cosine, not secant.

The **inverse** functions run the other way, recovering the angle from the ratio. But $\sin$, $\cos$, and $\tan$ each take every value infinitely often, so an inverse can exist only on a restricted domain, and the standard choices fix a single output range for each.

Function	Domain	Range
$\arcsin x$	$-1 \leq x \leq 1$	$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
$\arccos x$	$-1 \leq x \leq 1$	$0 \leq y \leq \pi$
$\arctan x$	all real x	$-\frac{\pi}{2} < y < \frac{\pi}{2}$

Because each inverse is a reflection of its restricted parent in the line $y = x$ (Ch. 5), the graphs are the familiar curves turned on their sides: arcsine climbs steeply from $(-1, -\frac{\pi}{2})$ to $(1, \frac{\pi}{2})$, arccosine falls from $(-1, \pi)$ to $(1, 0)$, and arctangent flattens out towards the horizontal asymptotes $y = \pm\frac{\pi}{2}$, the old vertical asymptotes of tangent lying down.

**EXAMPLE 9.7**

Find the exact values of (a) $\arcsin(-\frac{1}{2})$, (b) $\arccos(-\frac{\sqrt{3}}{2})$, (c) $\arcsin(\sin \frac{5\pi}{6})$.

- (a) The angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ whose sine is $-\frac{1}{2}$ is $-\frac{\pi}{6}$. Not $\frac{11\pi}{6}$ and not $\frac{7\pi}{6}$: those solve the equation but lie outside arcsine's range.
- (b) The angle in $[0, \pi]$ whose cosine is $-\frac{\sqrt{3}}{2}$ is $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Arccosine handles negatives by reaching into the second quadrant.
- (c) $\sin \frac{5\pi}{6} = \frac{1}{2}$, and $\arcsin \frac{1}{2} = \frac{\pi}{6}$. The composition returns $\frac{\pi}{6}$, not $\frac{5\pi}{6}$: arcsine can only answer with angles from its own range.

CAUTION $\arcsin(\sin x)$ does not always return x . Because arcsin only outputs angles in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, feeding it $\sin(2)$ gives $\pi - 2$, not 2. The restricted range is doing exactly its job.

YOUR TURN 9.4 Reciprocal and Inverse Functions HL

22. Evaluate $\sec \frac{\pi}{3}$. [1]

23. Evaluate $\csc \frac{\pi}{2}$. [1]

24. Evaluate $\cot \frac{\pi}{4}$. [1]

25. State the range of $\arcsin x$. [1]

26. Evaluate $\arcsin \frac{1}{2}$. [1]

27. Evaluate $\arccos 0$. [1]

28. Simplify $\sec^2 \theta - \tan^2 \theta$. [1]

9.5 Solving Trigonometric Equations

An equation like $\sin x = \frac{1}{2}$ has not one solution but infinitely many, because the wave crosses any horizontal line over and over. The calculator gives one angle; the unit circle gives the rest. Find the **principal** value, then use the symmetry of the circle to find every other solution in the required interval.

EXAMPLE 9.8

Solve $\sin x = \frac{1}{2}$ for $0 \leq x \leq 2\pi$.

The principal value is $x = \frac{\pi}{6}$. Sine is also positive in the second quadrant, where the partner angle is $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Both lie in the interval, so

$$x = \frac{\pi}{6} \quad \text{or} \quad x = \frac{5\pi}{6}.$$

Always ask which *other* quadrant shares the sign; that second angle is the one candidates most often miss.

TIP On a calculator paper, always check your solution count graphically: plot both sides and count the intersections in the interval. If your algebra found three solutions and the screen shows four, the screen is right.

When the argument is not plain x but $2x$ or $3(x-4)$, substitute for the whole argument and, crucially, transform the interval along with it. A doubled argument sweeps the circle twice as fast, so it collects twice as many solutions.

EXAMPLE 9.9

Solve $\sin 2x = \frac{1}{2}$ for $0 \leq x \leq 2\pi$.

Let $u = 2x$. As x runs over $[0, 2\pi]$, u runs over $[0, 4\pi]$: two full turns. In the first turn, $\sin u = \frac{1}{2}$ at $u = \frac{\pi}{6}$ and $\frac{5\pi}{6}$; the second turn repeats them 2π later, at $\frac{13\pi}{6}$ and $\frac{17\pi}{6}$. Halving each u recovers x :

$$x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}.$$

Four solutions, not two. Solving in u over the *enlarged* interval first, and only then dividing, is what protects you from losing the second pair.

Using identities

An equation mixing $\sin 2x$ with $\sin x$, or $\cos 2x$ with $\cos x$, cannot be solved as it stands. An identity rewrites everything in terms of a single function, turning the equation into an algebraic one, usually a quadratic.

EXAMPLE 9.10

Solve $\cos 2x + \cos x = 0$ for $0 \leq x \leq 2\pi$.

Choose the form of $\cos 2x$ that gives only cosines, $\cos 2x = 2\cos^2 x - 1$:

$$2\cos^2 x - 1 + \cos x = 0 \implies 2\cos^2 x + \cos x - 1 = 0.$$

This factors as $(2\cos x - 1)(\cos x + 1) = 0$, so $\cos x = \frac{1}{2}$ or $\cos x = -1$. From the circle,

$$x = \frac{\pi}{3}, \frac{5\pi}{3} \quad \text{or} \quad x = \pi.$$

Picking the right form of the double-angle identity, the one matching the other terms, is the whole art of these problems.

Enrichment: the wave form $a \sin \theta + b \cos \theta$

This last idea sits just beyond the AA syllabus, but it is too beautiful, and too useful, to leave out. A sum of a sine and a cosine of the same angle is itself a single shifted sine wave: the compound-angle identity, run backwards. Writing it as $R \sin(\theta + \alpha)$ makes both its maximum and its equation solvable at a glance.

KEY $a \sin \theta + b \cos \theta = R \sin(\theta + \alpha)$, where $R = \sqrt{a^2 + b^2}$ and $\tan \alpha = \frac{b}{a}$. The maximum value is R , the minimum $-R$.

In the same spirit, when a problem asks for *all* solutions rather than those in a fixed interval, the answer is a family generated by the period: $\sin x = k$ has general solution $x = \arcsin k + 2n\pi$ or $x = (\pi - \arcsin k) + 2n\pi$, for every integer n . The AA syllabus never asks for general solutions in an exam, but seeing the infinite family explains why examiners must always fence you into an interval.

YOUR TURN 9.5 Solving Trigonometric Equations

29. Solve $\sin x = \frac{1}{2}$ for $0 \leq x \leq 2\pi$. [2]

30. Solve $\cos x = 0$ for $0 \leq x \leq 2\pi$. [2]

31. Solve $\tan x = 1$ for $0 \leq x \leq 2\pi$. [2]

32. Solve $\sin x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$. [2]

33. Solve $2 \cos x = \sqrt{3}$ for $0 \leq x \leq 2\pi$. [2]

34. Solve $\cos x = 1$ for $0 \leq x \leq 2\pi$. [1]

35. Solve $\sin 2x = 0$ for $0 \leq x \leq 2\pi$. [2]

Reflections

IM Trigonometry began as astronomy. Babylonian and Greek observers needed to predict the wandering of the planets, and the sine was born as a tool for the geometry of the celestial sphere. Its greatest early triumph was in the Islamic world and in India, where astronomers tabulated sines to many places to fix the times of prayer and the direction of Mecca, and it returned to Europe through Latin translations that gave us the very word “sine”. For two thousand years, the functions of this chapter were the mathematics of the heavens.

TOK A sine wave is periodic: it repeats forever, identical in every cycle. Yet it is built from the endlessly irrational π and takes irrational values at almost every point. How can something so regular be made of something so unruly? And when the physicist Joseph Fourier proved that *any* repeating signal, however jagged, is a sum of pure sine waves, he suggested that the smooth circle hides inside every square wave and every spoken word. Is the sinusoid a human tool, or the alphabet in which nature writes all its rhythms?

RW Because a sine wave is what steady rotation looks like from the side, it models every oscillation in science and engineering. Alternating current rises and falls as a sine as the generator spins; sound is pressure oscillating sinusoidally, and its pitch is the wave's frequency; light, radio, and every signal your phone sends and receives are sinusoids. Fourier analysis, decomposing a signal into its sine components, is how music is compressed, images are stored, and earthquakes and heartbeats are read.

EXT The compound-angle identities hide a deeper unity, revealed by the complex numbers of Ch. 18. Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ turns the awkward addition rules for sine and cosine into the single, effortless law of exponents $e^{iA} e^{iB} = e^{i(A+B)}$. All the identities of this chapter are one fact about multiplying complex numbers on the unit circle, seen through the

fog of real notation.

End-of-Chapter Problems — Chapter 9: Trigonometric Functions

1. Find the exact values of (a) $\sin \frac{2\pi}{3}$; (b) $\cos \frac{2\pi}{3}$; (c) $\tan \frac{2\pi}{3}$. [3]
2. Given $\cos \theta = -\frac{3}{5}$ with θ in the second quadrant:
- (a) find $\sin \theta$; [2]
(b) find $\tan \theta$; [1]
(c) find $\sin 2\theta$. [2]
3. Prove the identity $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$. [4]
4. Solve $2 \sin^2 x + \sin x - 1 = 0$ for $0 \leq x \leq 2\pi$. [5]
5. Consider $f(x) = 3 \sin(2x) + 1$.
- (a) State the amplitude and period. [2]
(b) State the maximum and minimum values. [2]
6. Solve $\cos 2x = \cos x$ for $0 \leq x \leq 2\pi$. [5]
7. Prove that $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$. [4]
8. Find the exact value of $\tan 75^\circ$ using a compound angle. [4]
9. Solve $\sqrt{3} \tan x = 1$ for $0 \leq x \leq 2\pi$. [3]
10. The depth of water in a harbour is $d(t) = 5 + 3 \sin\left(\frac{\pi t}{6}\right)$ metres, t hours after midnight.
- (a) State the maximum and minimum depths. [2]
(b) State the period. [1]
11. Express $3 \sin x + 4 \cos x$ in the form $R \sin(x + \alpha)$ with $R > 0$.
- (a) Find R and α . [3]
(b) State the maximum value. [1]
12. Given $\tan \theta = 2$ with θ acute:
- (a) find $\sin \theta$ and $\cos \theta$; [2]
(b) find $\sin 2\theta$. [2]

13. Solve $2 \cos^2 x = 1$ for $0 \leq x \leq 2\pi$. [4]
14. Sketch $y = \cos x - 1$ for $0 \leq x \leq 2\pi$, stating the maximum, the minimum, and the intercepts. [4]
15. Given the identity $1 + \tan^2 \theta = \sec^2 \theta$, solve $\sec^2 x = 4$ for $0 \leq x \leq \frac{\pi}{2}$. [4]
16. Solve $\sin 2x = \sin x$ for $0 \leq x \leq 2\pi$. [5]
17. The graph of $y = a \sin(bx) + d$ has maximum 7, minimum 1, and period π . Find a , b and d . [4]
18. Prove that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. [5]
19. Solve $\tan^2 x = 3$ for $0 \leq x \leq 2\pi$. [4]
20. Evaluate, giving exact values: (a) $\arctan 1$; (b) $\arcsin(-\frac{1}{2})$; (c) $\arccos(-1)$. [3]
21. A seat on a Ferris wheel has height $h(t) = 10 - 8 \cos\left(\frac{\pi t}{15}\right)$ metres, t seconds after start.
- (a) State the minimum and maximum heights. [2]
- (b) Find the time for one full revolution. [2]
22. Solve $4 \sin^2 x - 3 = 0$ for $0 \leq x \leq 2\pi$. [4]
23. Prove that $\cos^4 \theta - \sin^4 \theta = \cos 2\theta$. [4]
24. Solve $2 \sin x \cos x = \frac{1}{2}$ for $0 \leq x \leq 2\pi$. [4]

Challenge Questions

25. Solve $\sin x + \cos x = 1$ for $0 \leq x \leq 2\pi$. [6]
26. Prove that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$. [5]
27. Solve $\cos 2x + 3 \sin x = 2$ for $0 \leq x \leq 2\pi$. [6]
28. Find the maximum value M of $5 \sin x + 12 \cos x$, and the value of x in $[0, 2\pi]$ at which it occurs. [5]

29. Prove that $\frac{1 + \sin 2\theta}{\cos 2\theta} = \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}$. [6]

